

LESSON 2-4 • SESSION 2 • PRACTICE • FORM B

Comparing Box Plots

Learning target – I can compare two groups by reading the medians and box widths of their box plots. · 6.SP.B.4

REMEMBER

- Put both box plots on the **same scale**.
- Compare **medians** for center: which group is typically higher, and by how much?
- Compare **box widths** ($Q3 - Q1$) for spread: the narrower box is more **consistent**.

WORKED EXAMPLE

Trail X: Q1 18, median 20, Q3 22. Trail Y: Q1 19, median 26, Q3 33 (minutes).

Step 1 Medians: $26 - 20 = 6$, so Y is typically 6 min longer.

Step 2 Box widths: X is $22 - 18 = 4$; Y is $33 - 19 = 14$.

Answer Y is longer; X is more consistent

WORDS center (*centro*) · spread (*dispersión*) · consistent (*constante*)

PRACTICE

show your work · label units

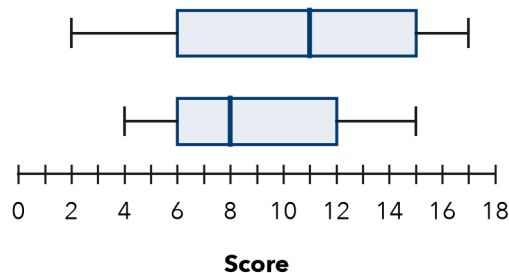
1. **Two Quizzes.** The box plots show scores on two quizzes.

Hint The line inside each box is the median.

- a. Higher median: _____
- b. By how many points: _____
- c. Wider box: _____

Quiz 2

Quiz 1



2. **Two Teams.** Team Gold points: 35, 41, 38, 47, 44, 49, 43. Team Green points: 30, 45, 36, 52, 40, 33, 50. Complete the table. Which team was more consistent?

Hint The narrower box is the more consistent team.

Team	Min	Q1	Median	Q3	Max
Gold					
Green					

More consistent _____

3. **Same Median.** Two plots have a median of 46. Box C runs from 44 to 49. Box D runs from 36 to 58.

- a. Width of box C: ____
- b. Width of box D: ____

4. **More Consistent?** Plot P has $Q1 = 31$ and $Q3 = 40$. Plot Q has $Q1 = 28$ and $Q3 = 45$. Which middle half is more consistent?

Hint Find each box width: $Q3 - Q1$.

☐ P ☐ Q

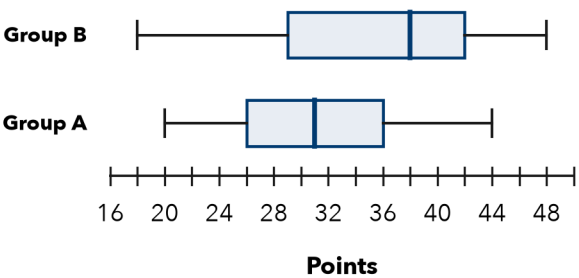
Widths _____

5. **Equal Spread?** Plot M runs from 10 to 80 with a box from 40 to 55. Plot N runs from 35 to 60 with a box from 44 to 50. Priya says they are equally spread. Is she right? Compare the spans and the boxes.

☐ Yes, equal spread ☐ No, M is more spread out ☐ No, N is more spread out

6. **Two Groups.** Which group has the higher median, and by how much?

Group; difference _____

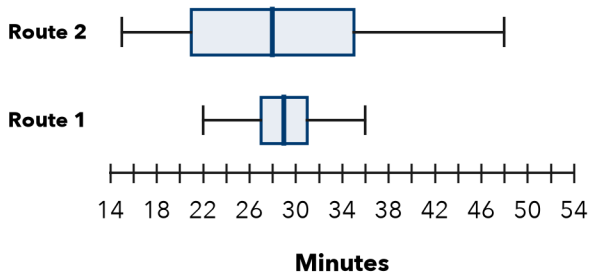


7. **Matching Boxes.** Plot W runs from 5 to 65. Plot Z runs from 22 to 41. Both boxes run from 25 to 35. Keisha says they show the same spread.

- a. Whole span of W (max – min): _____
- b. Whole span of Z: _____
- c. Is Keisha right? Explain.

8. **Two Bus Routes.** Travel times (minutes) for two routes.

- a. Higher median: _____
- b. More predictable (use box widths): _____
- c. A rider must not be late. Which route? Why?



FIND THE ERROR

explain the mistake, then fix it

9. **Dmitri's Work.** Comparing two box plots, Dmitri wrote: "Plot E has a longer right whisker, so Plot E has the greater median." Explain the mistake.

TEST PRACTICE

choose the best answer

10. A wider box means the middle half of the data is –

- A more spread out
- B closer together
- C higher
- D lower

11. A box plot has min 20 and max 45. Its whole span (max – min) is –

- A 65
- B 20
- C 25
- D 45

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

19 R	3 C	7 P	65 Q	60 A	Route 2 X	Gold O	Route 1 E	22 M	Green Z
#1b	#2	#3b	#6	#7a	#7b	#8b			

CHECK YOURSELF

- ☐ I can do this on my own
- ☐ I need a little help
- ☐ I have a question about #____